

Minor- II

School of Mathematics and Statistics
University of Hyderabad

September 30, 2026

Duration: 60 minutes
Maximum Score: 20 points

Instructor: Dharmendra Kumar
Course code: ST 402

Instructions: You may use results proven in the lectures; however, answers without justification will receive a score of zero.

1. Let $T : V \longrightarrow V$ be a linear operator on finite - dimensional vector space V . Then show that the following are equivalent:

(i) T is $1 - 1$.

(ii) T is onto.

[4]

2. Let $\mathcal{P} := \mathbb{R}[x]$ be vector space of polynomials over $\mathbb{R}$. Consider the following linear operators:

DERIVATIVE: $D(a_0 + a_1x + \cdots + a_nx^n) = a_1 + 2a_2x + \cdots + na_nx^{n-1}$

ANTIDERIVATIVE: $A(a_0 + a_1x + \cdots + a_nx^n) = a_0x + \frac{1}{2}a_1x^2 + \cdots + \frac{1}{n+1}a_nx^{n+1}$

(a) Show that D is onto but not $1 - 1$.

[4]

(b) Show that A is $1 - 1$ but not onto.

[4]

(c) Does above statements (a) and (b) contradict question 1 ?. Please justify it !.

[4]

3. Let $T : V \longrightarrow V$ be a linear operator on finite - dimensional vector space V . Let B_1 and B_2 be two ordered basis of V . Show that $[T]_{B_1}$ and $[T]_{B_2}$ are similar matrix.

[4]

All the best !